Artifact-Governed Black-Box Command-to-Motion Response Modeling

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Abstract

Closed-source legged robots present a deployment challenge: navigation systems issue velocity commands through manufacturer-provided SDK interfaces, but the mapping from commanded velocity to actual robot motion is opaque to the user. This paper presents an offline, artifact-governed pipeline for characterizing the black-box command-to-motion response relationship of a closed-source quadruped. The pipeline converts real measurement artifacts into a schema-validated velocity response dataset, produces conservative response predictions with categorical uncertainty and reliability labels, and translates prediction evidence into advisory navigation-risk assessments with explicit warning metadata. Evaluated on sparse forward-velocity evidence from a single K1 robot, the current implementation produces five dataset records (four numeric, one qualitative-only), five response predictions with uncertainty labels, and five advisory risk assessments across three risk levels. The work provides an evidence-governed foundation for studying deployment-layer reliability in closed-source legged robots and identifies the experiments required before navigation outcome claims can be evaluated.

1 Introduction

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[Section content placeholder.]

2 Method

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Table 1: Method Stage I/O Contract

Stage	Input	Output	Script	Non-Goals
1. Measurement	K1 field logs	Profile v0	M7/M8 workflow	Not compensation-ready
2. Dataset	Measurement v0 + schema	Dataset v1	dataset builder	No fabrication
3. Model	Dataset v1	5 predictions	model script	Not calibrated
4. Risk mapping	Predictions	5 assessments	risk script	No control
5. Evaluation	All outputs	Report + governance	eval script	Not pub-ready

3 Experiments

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Experiments placeholder.

Table 2: Current Evaluation Metrics

Metric	Available	Value
Dataset records	yes	5 (4 numeric, 1 qualitative)
Response predictions	yes	5
Risk assessments	yes	5 (critical=1, high=2, medium=2)
Advisory warnings	yes	5
Exact-source MAE	sanity check only	0.0 m/s
Held-out prediction error	no	—
Collision / near-miss / success	no	—
Calibrated uncertainty	no	—
Multi-surface generalization	no	—

4 Discussion

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Table 3: Evidence Boundary and Claim-Upgrade Requirements

Claim Type	Current Status	Required Evidence
Pipeline existence	supported (structural)	Already met
Dataset construction	supported (structural)	Already met
Model predictive quality	not supported	Held-out evaluation on repeated trials
Risk warning usefulness	not supported	Navigation outcome trials
Navigation safety improvement	prohibited	Trials + collision/success metrics
Calibrated uncertainty	prohibited	Repeated trials + calibration protocol
Generalization	prohibited	Multi-surface, multi-session data
Compensation readiness	prohibited	All above + offline validation
Safe command adapter	prohibited	All above + safety validation

5 Conclusion

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Conclusion placeholder.

6 Current Evidence and Missing-Evidence Boundary (Figure 3 — optional)

[Figure: current_evidence_gap_v1.md – not rendered]

7 Appendix Tables

The following tables are recommended for appendix/supplement inclusion per venue page limits:

- Table A1: Numeric Traceability ([path])
- Table B1: Citation Audit ([path])
- Table C1: Terminology Consistency ([path])
- Table D1: Claim-Upgrade Evidence Matrix ([path])

Full appendix table pack: [path].

8 Future Experiment Protocol

The M20 future experiment protocol ([path]) defines 4 experiment tiers with 35 metrics. M21/M21.1 data collection pack ([path]) provides templates and checklists. No experiments have been executed.

9 Evidence Gaps

- No real navigation outcome evidence.
- No held-out command evaluation.
- Single robot, single surface, single session.
- Sparse command grid (5 forward-velocity points).
- No calibrated uncertainty.
- No collision, near-miss, or success-rate metrics.